

UNIT IV

LASERS

B.Tech ECE / VLSI — Engineering Physics

Light Amplification by Stimulated Emission of Radiation

□ MIND-BLOWING LASER FACTS

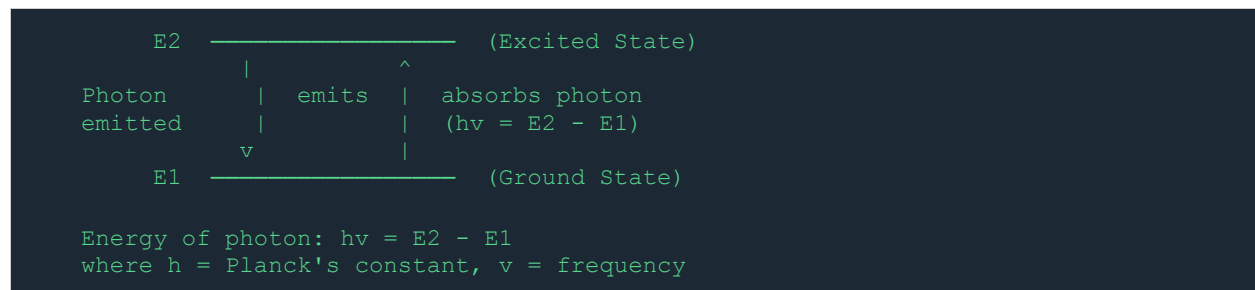
- Lasers cut diamonds — the hardest natural material on Earth!
- Lasers perform LASIK eye surgery, reshaping the cornea in under 30 seconds
- The entire internet runs on laser-powered fiber optics — right now!
- Chip makers use EUV lasers to write circuit patterns at 3 nm scale
- LiDAR in self-driving cars fires millions of laser pulses per second
- The red dot at a supermarket barcode scanner? That's a He-Ne laser!

TOPIC 1 — INTERACTION OF LIGHT WITH MATTER

Before lasers can make sense, we need to understand what happens when light meets an atom. There are exactly THREE things that can happen — no more, no less.

The Energy Level Ladder

Atoms exist in discrete energy states — think of them as rungs on a ladder. An atom normally sits at the lowest rung (ground state). To move up, it must absorb a precise amount of energy. It cannot sit between rungs.



Process 1 — Absorption

When a photon with energy equal to $(E2 - E1)$ strikes a ground-state atom, the atom absorbs the photon and jumps to the excited state. The photon is destroyed.

- Atom in ground state ($E1$) + photon ($h\nu$) → Atom in excited state ($E2$)
- Requires exact photon energy: $h\nu = E2 - E1$
- The photon simply vanishes — absorbed by the atom
- ECE Link: Photodetectors and solar cells exploit absorption

Process 2 — Spontaneous Emission

An excited atom is unstable. After a very short time (nanoseconds to microseconds), it spontaneously falls back to the ground state and emits a photon on its own, without any external trigger.

- Atom in excited state ($E2$) → Atom in ground state ($E1$) + photon ($h\nu$)
- Random direction, random phase, random polarization — INCOHERENT light
- Excited state lifetime: typically 10^{-8} seconds
- Example: Stars, fire, incandescent bulbs — ALL emit by spontaneous emission
- ECE Link: LEDs emit by spontaneous emission

Process 3 — Stimulated Emission (The Magic Behind Lasers!)

This is Einstein's genius insight from 1917. If a photon hits an ALREADY EXCITED atom, instead of being absorbed, it stimulates the atom to emit a SECOND identical photon!

- Photon ($h\nu$) + Excited atom (E_2) $\rightarrow$ 2 identical photons + Ground state atom (E_1)
- Output photon is IDENTICAL to input: same direction, phase, frequency, polarization
- This is AMPLIFICATION: 1 photon in $\rightarrow$ 2 photons out!
- The light produced is COHERENT — all photons march in step

□ LEMON-SIMPLE ANALOGY — The Sneeze Room

Spontaneous Emission: One person in a quiet room suddenly sneezes randomly in any direction. No pattern, no coordination.

Stimulated Emission: One person sneezes, and it instantly triggers EVERYONE in the room to sneeze at the SAME TIME, in the SAME direction, at the SAME intensity. That coordinated sneeze-wave is laser light!

Property	Spontaneous Emission vs Stimulated Emission
Trigger	Spontaneous (no trigger) Requires incident photon
Direction	Random Same as incident photon
Phase	Random Same as incident photon
Coherence	Incoherent Coherent
Example	LED, Sun, Bulb LASER

✓ QUICK CHECK

1. What are the three ways light can interact with matter?
2. Why is stimulated emission called amplification?
3. What makes stimulated emission different from spontaneous emission?
4. Why are LEDs incoherent and lasers coherent?

TOPIC 2 — EINSTEIN'S A AND B COEFFICIENTS

□ Historical Moment

In 1917, Albert Einstein published a paper predicting stimulated emission — THIRTEEN YEARS before quantum mechanics was even complete! He derived it purely from thermodynamic arguments. This is the theoretical foundation of every laser on Earth.

Einstein described the rates of the three light-matter processes using two fundamental constants — now called the Einstein A and B coefficients.

Einstein A Coefficient (Spontaneous Emission Rate)

- Symbol: A_{21} (or just A)
- Definition: Probability per second that an excited atom spontaneously emits a photon
- Units: s^{-1} (per second)
- Physical meaning: Higher A = faster spontaneous decay = shorter excited state lifetime
- Relationship to lifetime: $A = 1/\tau$ where τ is the mean lifetime of the excited state

Einstein B Coefficient (Stimulated Emission / Absorption Rate)

- Symbol: B_{21} (stimulated emission), B_{12} (absorption)
- Definition: Probability of stimulated emission per unit radiation energy density
- Units: $m^3 J^{-1} s^{-2}$ (complex units, often just noted as B)
- Rate of stimulated emission = $B_{21} \times u(\nu)$ where $u(\nu)$ = radiation energy density
- For non-degenerate levels: $B_{12} = B_{21}$ (absorption and stimulated emission equally probable!)

The Key Relationship: A/B Ratio

Einstein derived the relationship between A and B by requiring consistency with Planck's blackbody radiation formula at thermal equilibrium:

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$$A / B = 8 \pi h \nu^3 / c^3$$

Where:

- A** = Einstein A coefficient (spontaneous emission rate)
- B** = Einstein B coefficient (stimulated emission rate)
- h** = Planck's constant = $6.626 \times 10^{-34} \text{ J}\cdot\text{s}$

ν = frequency of radiation (Hz)
 c = speed of light = 3×10^8 m/s

Physical Meaning of the A/B Ratio

This equation tells us something profound: A/B increases as ν^3 . This means:

- At OPTICAL frequencies (high ν): A/B is HUGE → Spontaneous emission dominates
- At RADIO frequencies (low ν): A/B is small → Stimulated emission more competitive
- At thermal equilibrium, spontaneous emission ALWAYS wins over stimulated emission
- To make a laser: we MUST create non-equilibrium conditions (population inversion!)

□ ECE/VLSI Application

The A/B ratio explains why making a laser at shorter wavelengths (UV, X-ray) is so much harder than infrared lasers. For EUV lithography (used in chip-making at 3nm), achieving lasing at 13.5 nm requires extreme plasma sources. The A coefficient at that wavelength is astronomically high, making population inversion incredibly difficult to sustain.

✓ QUICK CHECK

1. What does the A coefficient represent physically?
2. If A is large, what happens to the excited state lifetime?
3. Write the Einstein A/B relationship formula
4. Why does higher frequency make lasing harder to achieve?

TOPIC 3 — SPONTANEOUS VS STIMULATED EMISSION

Detailed Comparison

Property	Spontaneous Emission	Stimulated Emission
Trigger	None — occurs randomly	Incident photon required
Rate	A_{21} (constant)	$B_{21} \times u(\nu)$ — depends on field
Phase	Random	Same as incident photon
Direction	All directions (isotropic)	Same as incident photon
Polarization	Random	Same as incident photon
Frequency	Same $(E_2 - E_1)/h$	Same as incident photon
Coherence	INCOHERENT	COHERENT
Photon count	1 photon out	2 identical photons out
Example device	LED, Sun, Lamp	LASER

Why Coherence Matters in ECE/VLSI

Coherent light (same phase, direction, frequency) is what makes lasers special for engineering. Here's why:

- Fiber optic communication: Coherent laser light carries data with minimal dispersion over thousands of km
- Holography: Requires coherent light to record 3D interference patterns
- Interferometry: Used for ultra-precise measurements in VLSI (measuring chip features at nm scale)
- Optical lithography: EUV lasers write circuits with precision impossible without coherent light

Stimulated Emission — The Amplification Mechanism

The chain reaction of stimulated emission is what makes laser amplification possible:

```
1 photon
  |
  v  [excited atom]
+-----+
| STIM. | --> 2 photons (identical)
| EMISS.|
```

```

+-----+
|
|  v   [excited atom]
+-----+
| STIM. | --> 4 photons (identical)
| EMISS.|
+-----+
|
|  v   [and so on...]
EXPONENTIAL AMPLIFICATION! (if population inverted)

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Rate of Stimulated Emission = $B_{21} \times u(\nu) \times N_2$

Rate of Absorption = $B_{12} \times u(\nu) \times N_1$

Rate of Spontaneous Emission = $A_{21} \times N_2$

Net amplification occurs when: $N_2 > N_1$ (Population Inversion!)

N_1 = atoms in lower level, N_2 = atoms in upper level

✓ QUICK CHECK

1. What is the key property that distinguishes stimulated emission photons from spontaneous ones?
2. Write the expression for the rate of stimulated emission
3. Why can't we get laser action without population inversion?
4. Name two ECE applications that specifically require coherent light

TOPIC 4 — POPULATION INVERSION

Normal vs Inverted Population

In thermal equilibrium, atoms obey the Boltzmann distribution — the majority are in the ground state. This is the natural, stable condition.

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Boltzmann Distribution: $N_2 / N_1 = \exp(-(E_2 - E_1) / kT)$

At room temperature ($T = 300 \text{ K}$), for optical transitions:

$N_2/N_1 \sim 10^{-40}$ (essentially ALL atoms are in ground state!)

For laser action, we NEED: $N_2 > N_1$ (Population Inversion)

NORMAL (Thermal Equilibrium)

E2 — ● —
(few atoms)

E1 — ●●●● —
(many atoms)

Absorption DOMINATES
--> No laser action

POPULATION INVERSION
(Non-equilibrium)

E2 — ●●●● —
(MANY atoms)

E1 — ● —
(few atoms)

Stimulated Emission DOMINATES
--> LASER action possible!

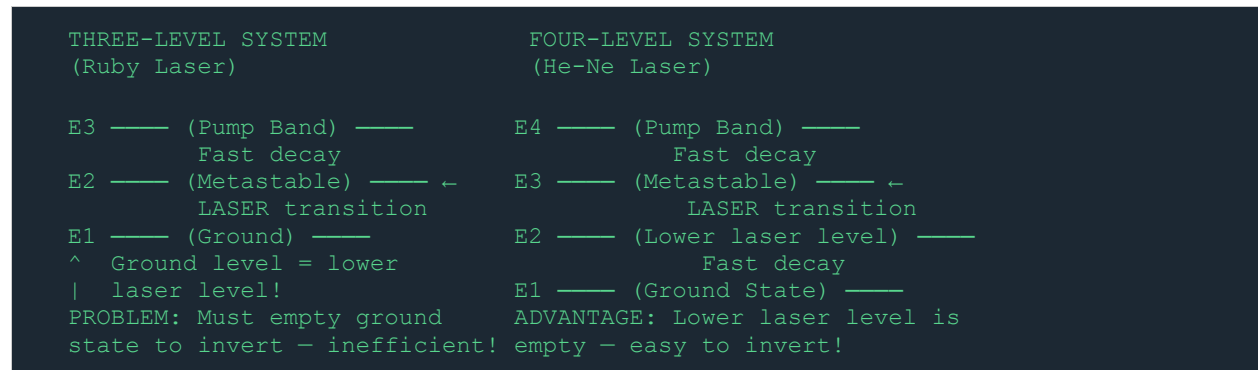
How to Achieve Population Inversion — PUMPING

Since population inversion is against thermal equilibrium, we must continuously supply energy to maintain it. This process is called PUMPING.

Pumping Method	How It Works	Used In
Optical Pumping	Bright flash lamp or another laser excites atoms	Ruby laser, Nd:YAG
Electrical Discharge	Electric current excites gas atoms by collisions	He-Ne laser, CO2 laser
Chemical Pumping	Exothermic chemical reaction provides energy	Chemical lasers
Injection (p-n junction)	Forward biased p-n junction injects carriers	Semiconductor diode lasers

Three-Level vs Four-Level Laser Systems

The number of energy levels involved determines how efficiently population inversion can be achieved:



Three-Level System (Ruby)	Four-Level System (He-Ne, Nd:YAG)
Lower laser level = ground state	Lower laser level is above ground state
Must excite >50% of ALL atoms	Only need to populate upper level
Requires intense pumping	Less pumping needed
Less efficient	More efficient
Pulsed operation only	Continuous wave (CW) possible
$N_2 > N_1$ harder to achieve	$N_2 > N_1$ achieved easily

✓ QUICK CHECK

1. What is population inversion? Why is it a non-equilibrium condition?
2. Write the Boltzmann formula and use it to explain why $N_2 \ll N_1$ normally
3. Name four methods of pumping
4. Why is the four-level system more efficient than three-level?

TOPIC 5 — COMPONENTS OF A LASER

Every laser, from a tiny diode to a giant industrial cutter, has exactly three essential components. Think of them as: the fuel, the engine, and the echo chamber.

Component 1 — Active Medium (The Fuel)

The active medium is the material that actually amplifies light through stimulated emission. It must have a metastable state where population inversion can be achieved.

- Also called: gain medium or lasing medium
- Can be solid (Ruby crystal, Nd:YAG glass), gas (He-Ne, CO₂), liquid (dye lasers), or semiconductor (GaAs diode)
- Must have an energy level structure suitable for population inversion
- The transition between levels determines the laser's output wavelength
- ECE: Semiconductor active media (GaAs, InGaAsP) are used in fiber-optic transceivers

Component 2 — Pumping Mechanism (The Engine)

The pumping mechanism supplies energy to create and sustain population inversion. Without continuous pumping, spontaneous emission would bring the atoms back to equilibrium.

- Optical pumping: Flash lamps or pump lasers irradiate the active medium
- Electrical discharge: Current through gas excites atoms by electron collisions
- Chemical reaction: Exothermic reactions create excited molecules directly
- Junction injection: Forward bias drives electrons into excited states in diode lasers

Component 3 — Optical Resonator / Cavity (The Echo Chamber)

Two mirrors surrounding the active medium trap photons and make them pass through the gain medium repeatedly. This is how a single photon becomes an intense beam.

```
100% Mirror      Active Medium      Partial Mirror (90-95%)
|M1| <-----> |M2|-----> LASER OUTPUT
| | <- photons bounce -> | |
| | <- gain on each pass -> | |
```

- M1 = Fully reflective (reflects 100%)
- M2 = Partially reflective (transmits 5-10% = output beam)
- Photons bounce back and forth, stimulating emission each pass
- Beam grows exponentially until losses = gain (threshold!)

- Full mirror (M1): Reflects 100% of light back into the medium
- Output coupler (M2): Partially reflective — lets ~5-10% escape as the laser beam

- Resonator length must satisfy: $L = n \times (\lambda/2)$ — standing wave condition
- Threshold condition: Gain per round trip must exceed all losses (absorption, scattering, mirror loss)

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Threshold Condition: $g(\nu) \geq \alpha + (1/2L) \times \ln(1/R_1R_2)$

Where:

$g(\nu)$ = gain coefficient of the medium

α = distributed loss (absorption, scattering)

L = length of active medium

R_1, R_2 = reflectivities of the two mirrors

□ ANALOGY — The Echo Chamber

Imagine you're in a room with perfect mirrors on two walls. You clap once (pump), and that sound bounces back and forth. Each time it passes through a microphone-amplifier in the room (active medium), it gets louder. Eventually, it's loud enough that a little sound leaks through one slightly imperfect wall — that's your laser output beam!

✓ QUICK CHECK

1. Name the three essential components of a laser
2. What is the role of the optical resonator?
3. Why does one mirror need to be partially reflective?
4. State the threshold condition for laser oscillation

TOPIC 6 — RUBY LASER

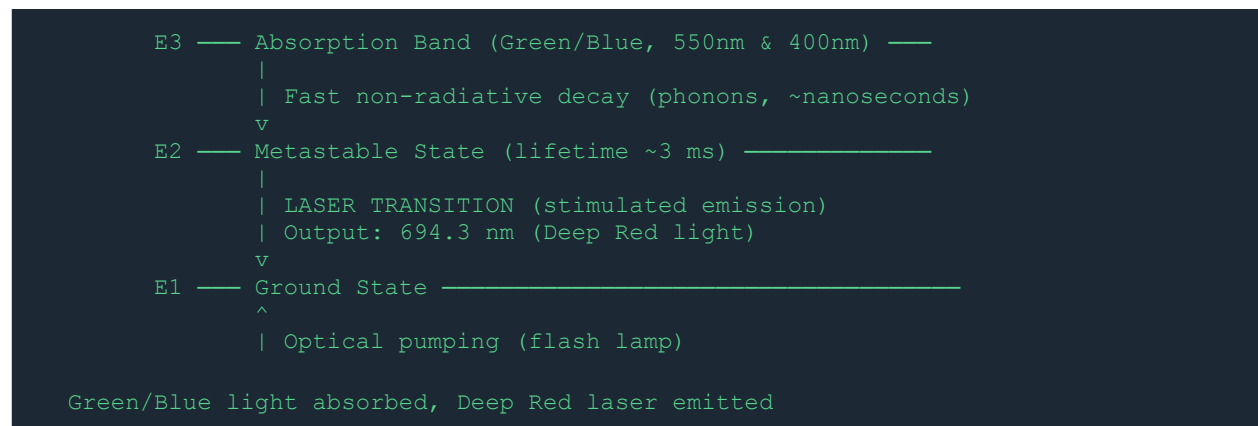
□ Historical Achievement

The Ruby Laser was the world's FIRST working laser, demonstrated by Theodore Maiman on May 16, 1960 at Hughes Research Laboratories. It changed history! The active medium was a synthetic ruby crystal powered by a xenon flash lamp.

Construction & Components

- Active medium: Ruby crystal (Al_2O_3 doped with Cr^{3+} chromium ions, ~0.05% by weight)
- The Cr^{3+} ions are the lasing species — aluminum and oxygen don't contribute
- Shape: Cylindrical rod, typically 5-20 cm long, 5-10 mm diameter
- Pumping: Xenon flash lamp wound helically around the ruby rod
- Mirrors: One end fully silvered (100%), other end partially silvered (~80%)
- Cooling: Water or air cooling required (high heat generation)

Energy Level Diagram — Three-Level System



Working Principle — Step by Step

1. Flash lamp fires: Intense white light from xenon flash lamp floods the ruby rod
2. Absorption: Cr^{3+} ions absorb green (~550nm) and blue (~400nm) photons, jumping to E3
3. Fast relaxation: Atoms quickly (nanoseconds) fall from E3 to E2 (metastable state) via non-radiative decay — no photon emitted, energy goes to crystal vibrations (phonons)
4. Metastable accumulation: E2 has a long lifetime (~3 ms) — atoms pile up here while ground state depletes
5. Population inversion: If pumping is intense enough, $N_2 > N_1$ — population inversion achieved!

6. Stimulated emission: Spontaneous photon triggers chain of stimulated emissions at 694.3 nm
7. Resonator buildup: Photons bounce between mirrors, growing in intensity
8. Output: Laser pulse exits through the partially reflective mirror

Output Characteristics

Parameter	Value / Description
Wavelength	694.3 nm (deep red, visible)
Output mode	Pulsed (not continuous wave)
Pulse duration	Milliseconds per flash
Peak power	Very high (MW range possible)
Efficiency	Low (~0.1%) — 3-level inefficiency
Beam color	Bright red — visible to naked eye

Limitations

- Low efficiency: Lower laser level is the ground state — must excite >50% of ALL Cr³⁺ ions for inversion
- Pulsed only: During the gap between flashes, atoms return to ground state — continuous wave is not possible
- High pumping threshold: Intense flash lamp needed — causes heating
- Bulky system: Requires cooling, heavy flash lamps, and high-voltage power supply

□ ECE/VLSI Applications

- Laser ranging and rangefinders (LIDAR predecessors)
- Pulsed laser deposition — thin film fabrication for semiconductor devices
- Tattoo removal and dermatology (high peak power pulses)
- Holography — early applications in 3D imaging

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Ruby laser output wavelength: 694.3 nm

Energy transition: E₂ --> E₁ (Cr³⁺ metastable to ground)

Metastable lifetime: $\tau \sim 3 \text{ ms}$

Pumping: Optical (xenon flash lamp, green/blue absorption)

System type: 3-level

✓ QUICK CHECK

1. Who built the first working laser and when?
2. What is the active medium in a ruby laser and what ions actually lase?
3. Draw and label the three-level energy diagram for the ruby laser
4. What wavelength does the ruby laser emit? Is it visible?
5. Why can the ruby laser only operate in pulsed mode?

TOPIC 7 — HELIUM-NEON (He-Ne) LASER

□ Engineering Marvel

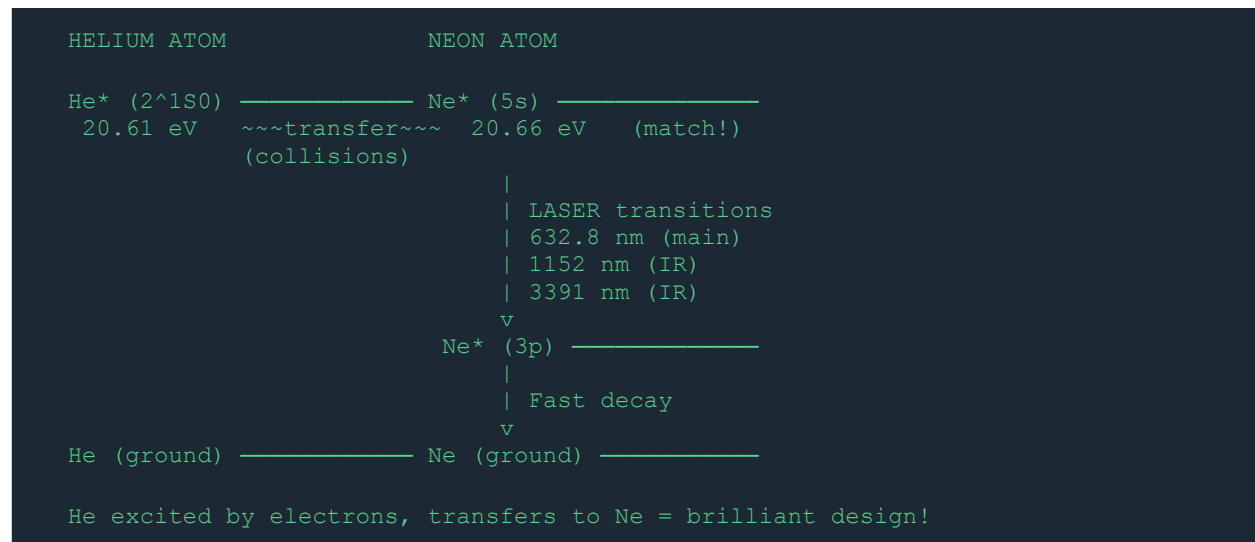
The He-Ne laser was the first CONTINUOUS WAVE (CW) laser and the first GAS laser, developed in 1960-61 by Ali Javan, William Bennett Jr., and Donald Herriott at Bell Labs. It is still one of the most widely used lasers in education and industry today!

Construction & Components

- Active medium: Mixture of Helium (He) and Neon (Ne) gases in ratio 10:1 or 85%:15%
- The NEON atoms actually lase — Helium is the energy transfer agent
- Gas tube: Sealed glass or quartz tube, 15-60 cm long, 2-7 mm diameter
- Pumping: Electrical discharge (RF or DC current through the gas)
- Electrodes: Anode and cathode create the discharge
- Mirrors: One fully reflective, one partially reflective (~99% reflectivity!)
- Pressure: Total gas pressure ~1 torr (0.133 kPa)

The Energy Transfer Secret — Why Helium?

This is the cleverest part of the He-Ne design! Helium and Neon happen to have energy levels that match almost perfectly — this is called resonance energy transfer.



Working Principle — Step by Step

9. Electrical discharge: High voltage applied — free electrons accelerate through the gas

10. Helium excitation: Fast electrons collide with He atoms, exciting them to a metastable state at ~20.61 eV
11. Energy resonance: Excited He* collides with ground-state Ne atoms — perfect energy match at ~20.66 eV transfers the excitation to Ne (4-level system's upper level)
12. Population inversion in Ne: Ne atoms accumulate in the 5s level (upper laser level) while 3p level (lower laser level) remains nearly empty
13. Stimulated emission: Ne atoms undergo stimulated emission: 5s → 3p, emitting 632.8 nm red photons
14. Fast decay: Ne atoms in 3p rapidly decay to the 3s level (preventing reabsorption)
15. Final return: Ne* in 3s returns to ground via spontaneous emission and wall collisions
16. Continuous operation: He returns to ground, gets re-excited — the cycle continues indefinitely!

Output Characteristics

Parameter	Value / Description
Main wavelength	632.8 nm (bright red, visible)
Other wavelengths	1152 nm, 3391 nm (infrared)
Output mode	Continuous Wave (CW)
Power output	0.5 mW to 50 mW (typical)
Efficiency	~0.01% — but very stable
Beam quality	Excellent — near-perfect Gaussian beam
Coherence length	Very long — meters to hundreds of meters
Lifetime	20,000+ hours of operation

Why He-Ne is a 4-Level System — Efficiency Advantage

- Upper laser level = 5s state of Ne (metastable, long lifetime)
- Lower laser level = 3p state of Ne (short lifetime — empties quickly!)
- Lower level is NOT the ground state — it's above ground state
- Result: Population inversion is easy to maintain at low pump power
- Continuous wave operation is natural because of this efficient level scheme

□ ECE/VLSI Applications

- Barcode scanners in supermarkets and logistics (the classic red dot!)
- Holography — coherent illumination for 3D recording
- Optical interferometry — measuring surface flatness in VLSI wafer inspection
- Fiber optic testing and alignment

- Laser alignment in construction and surveying
- Research and lab demonstrations — standard tool in optics labs worldwide

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He-Ne output wavelength: 632.8 nm (main visible line)

Energy transition: Ne 5s $\rightarrow$ 3p (stimulated emission)

He-Ne gas ratio: 85% He : 15% Ne (or 10:1)

Pumping: Electrical discharge (electron collisions)

System type: 4-level

Transfer: $\text{He}^* + \text{Ne} \rightarrow \text{He} + \text{Ne}^*$ (resonance energy transfer)

Metastable lifetime of Ne(5s): ~100 ns

✓ QUICK CHECK

1. What is the output wavelength of a He-Ne laser? What color is it?
2. Why is Helium added if Neon is the lasing medium?
3. Explain resonance energy transfer in He-Ne laser
4. Why can the He-Ne laser operate in CW mode while ruby cannot?
5. Draw the four energy levels for the He-Ne laser transition

TOPIC 8 — RUBY vs He-Ne: COMPLETE COMPARISON

Parameter	Ruby Laser	He-Ne Laser
Type	Solid-state laser	Gas laser
Active medium	Cr ³⁺ in Al ₂ O ₃ crystal	Neon gas (He as energy donor)
Pumping method	Optical (xenon flash lamp)	Electrical discharge
Energy levels	3-level system	4-level system
Output	Pulsed only	Continuous Wave (CW)
Wavelength	694.3 nm (deep red)	632.8 nm (bright red)
Efficiency	Low (~0.1%)	Very low but stable (~0.01%)
Power output	High peak power (MW)	Low average power (mW)
Threshold	High pump threshold	Low threshold
Beam coherence	Good	Excellent
Portability	Bulky, needs cooling	Compact, air-cooled
Cost	Higher	Lower
First demonstrated	1960 (T. Maiman)	1960-61 (Javan et al.)
Main applications	Ranging, pulsed cutting, dermatology	Barcode, holography, interferometry

BONUS — LASER APPLICATIONS IN ECE/VLSI

1. Semiconductor Lasers (Diode Lasers)

The backbone of modern optical communications. A forward-biased p-n junction creates population inversion at the junction — the entire internet runs on these devices.

- Active medium: Direct bandgap semiconductor (GaAs, InGaAsP, InGaN)
- Pumping: Injection of electrons and holes by forward bias current
- Wavelengths: 780 nm (CD), 650 nm (DVD), 405 nm (Blu-ray), 1310/1550 nm (fiber optic)
- Size: Smaller than a grain of rice — integrates on chips
- Efficiency: 30-70% — far superior to gas or solid-state lasers
- Application: SFP modules, optical transceivers, LIDAR, laser printing

2. EUV Laser Lithography

Extreme Ultraviolet (EUV) lithography is how modern chips at 3 nm, 5 nm, and 7 nm are made. TSMC, Samsung, and Intel all use ASML EUV machines.

- Wavelength: 13.5 nm (extreme ultraviolet, not really a laser — plasma source)
- Works by: Tin droplets hit by CO₂ laser → plasma emits EUV photons
- Resolution: Can pattern features as small as ~8 nm half-pitch
- Cost: Each EUV machine costs \$150-200 million USD
- Energy: 250+ kW of laser power needed
- Application: Manufacturing Apple M-series chips, Nvidia GPUs, modern processors

3. LiDAR (Light Detection and Ranging)

- Fires pulsed laser beams (905 nm or 1550 nm) and measures return time
- Time-of-flight gives precise 3D distance maps of surroundings
- Used in: Tesla Autopilot, Waymo self-driving cars, robotics, drones
- Resolution: Can detect objects smaller than 1 cm at 100+ meter range
- ECE link: Laser pulse generation, time-of-flight sensing, ADC/DAC systems

4. Laser Annealing in Semiconductor Fabrication

- Excimer laser (248 nm or 308 nm) rapidly heats and recrystallizes amorphous silicon
- Used in: LTPS (Low Temperature Poly-Silicon) for OLED display backplanes (your smartphone screen!)
- Advantage: Only heats the surface — no damage to underlying layers
- Also used for: Dopant activation, defect repair in ion-implanted layers

- ECE link: MOSFET threshold voltage control, contact resistance reduction

5. Optical Fiber Communications

- Semiconductor lasers at 1310 nm and 1550 nm carry data through glass fibers
- 1550 nm is optimal — matches the low-loss window of silica fiber (~ 0.2 dB/km)
- DWDM: 80+ channels of different wavelengths on a single fiber — terabit speeds
- Modulation: Lasers can be directly or externally modulated at 100+ Gb/s
- ECE link: Optical networking, 5G fronthaul, data center interconnects

MASTER FORMULA SHEET — UNIT IV LASERS

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1. Photon energy: $E = h\nu = hc/\lambda$
2. Einstein A/B ratio: $A/B = 8\pi h \nu^3 / c^3$
3. Excited state decay: $A = 1/\tau$ (τ = lifetime)
4. Rate of stim. emission: $R_{stim} = B_{21} \times u(\nu) \times N_2$
5. Rate of absorption: $R_{abs} = B_{12} \times u(\nu) \times N_1$
6. Boltzmann distribution: $N_2/N_1 = \exp(-(E_2-E_1)/kT)$
7. Population inversion: $N_2 > N_1$ (required for lasing)
8. Threshold condition: $g(\nu) \geq \alpha + (1/2L) \ln(1/R_1R_2)$
9. Ruby wavelength: 694.3 nm
10. He-Ne wavelength: 632.8 nm
11. Resonator condition: $L = n \times (\lambda/2)$

QUICK REVISION TABLE — ALL TOPICS

Topic	Key Concept	Must Remember
Absorption	Photon absorbed, atom excited to E_2	$h\nu = E_2 - E_1$
Spontaneous Emission	Random photon without trigger	Incoherent, all directions
Stimulated Emission	1 photon in, 2 identical out	Coherent, same direction/phase
Einstein A	Rate of spontaneous emission	$A = 1/\tau$
Einstein B	Rate of stim. emission/absorption	$B_{12} = B_{21}$ (non-degenerate)
A/B ratio	$A/B = 8\pi h \nu^3 / c^3$	Higher frequency = harder to lase
Population Inversion	$N_2 > N_1$ required	Against Boltzmann distribution
Pumping	Creates & sustains inversion	Optical, electrical, chemical, injection
3-level system	Lower level = ground state	Inefficient, pulsed (Ruby)
4-level system	Lower level above ground	Efficient, CW possible (He-Ne)
Active medium	Amplifies light (gain medium)	Must have metastable state
Optical resonator	Two mirrors trap photons	$M_1=100\%$, $M_2=\text{partial}$ (~95%)
Ruby laser	Cr^{3+} , 694.3nm, pulsed	3-level, optical pumped, 1960

He-Ne laser	Ne lases, He transfers energy	4-level, 632.8nm, CW, electrical
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END OF UNIT IV — LASERS

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